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Using Bid Rotation and Incumbency to Detect Collusion: A Regression Discontinuity Approach

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1 Big picture

- **Question.** Can regulators use standard bidding data to detect non-competitive behaviour when cartels allocate procurement contracts through bid rotation or incumbency rules?
- **Empirical challenge.** Bid rotation and incumbent advantages are classic red flags of collusion, but both can also arise competitively. A firm with high backlog may have high marginal costs, and an incumbent may have lower costs because of learning-by-doing, scale, or location.
- **Core idea.** Focus only on auctions in which the winner and the nearest loser are separated by a very small bid difference. Under competitive bidding, winning and losing by a tiny margin should be nearly as-if random conditional on the observed and unobserved state.
- **Running variable.** For bidder i in auction t , define

$$\Delta_{i,t} = b_{i,t} - \wedge b_{-i,t},$$

where $b_{i,t}$ is bidder i 's bid and $\wedge b_{-i,t}$ is the lowest competing bid. If $\Delta_{i,t} < 0$, bidder i wins; if $\Delta_{i,t} > 0$, bidder i loses.

- **Test statistic.** Compare bidder characteristics $x_{i,t}$, such as backlog or incumbency, just to the right and left of $\Delta_{i,t} = 0$. The discontinuity is

$$\beta = \lim_{\epsilon \downarrow 0} E[x_{i,t} \mid \Delta_{i,t} = \epsilon] - \lim_{\epsilon \uparrow 0} E[x_{i,t} \mid \Delta_{i,t} = \epsilon].$$

- **Null of competition.** Under competitive bidding with smooth residual demand, close winners and close losers should have the same distribution of observables. Therefore $\beta = 0$.
- **Collusion implication.** If a cartel rotates contracts to low-backlog firms, marginal winners should have lower backlog than marginal losers, so $\beta > 0$ when x is backlog. If a cartel gives priority to incumbents, marginal winners should be more likely to be incumbents than marginal losers, so $\beta < 0$ when x is incumbent status.
- **Theoretical contribution.** The paper proves when this close-bid randomization logic is valid in a dynamic procurement game. It introduces the notion of *competitively enforced* Markov perfect equilibrium and shows that, under this condition, close bids behave as if winning were random.
- **Empirical applications.** The paper applies the test to two settings: the Ohio school milk auctions studied by Porter and Zona and a large sample of municipal procurement auctions in Japan.
- **Main empirical findings.** In Ohio, known cartel firms show a discontinuity in incumbency around the winning threshold, but control firms do not. In Japan, close losers have higher backlog than close winners, and close winners are more likely to be incumbents, especially in high-bid auctions.
- **Policy takeaway.** Regulators can use close-bid discontinuities in covariates such as backlog and incumbency to screen for likely collusion without having to fully model costs or solve a structural procurement game.

2 Incremental contribution of the paper

2.1 Contribution relative to cartel-screening practice

- Existing antitrust screens often use suspicious patterns such as bid rotation, repeated winners, market division, identical bidding, or unusual bid spreads.
- The problem is that these screens are frequently descriptive. They flag patterns that are consistent with collusion but also consistent with competitive heterogeneity.
- This paper turns two common red flags—bid rotation and incumbency advantage—into formal regression discontinuity tests.

- The test asks whether marginal winners and marginal losers differ in pre-determined characteristics. Because both bid almost the same price, competitive cost-based explanations should not produce a discontinuity at zero.

Interpretation: The paper’s contribution is not merely another cartel screen. It provides a way to distinguish suspicious allocation patterns that reflect conduct from allocation patterns that merely reflect cost differences.

2.2 Contribution relative to empirical industrial organization

- Dynamic procurement markets are difficult because past wins affect future states through backlog, learning, capacity, reputation, and incumbency.
- A fully structural dynamic auction model would require modelling information, costs, state transitions, participation, and bidding strategies.
- The authors avoid this by exploiting a local discontinuity: conditional on bids being arbitrarily close, competitive winning is as-if random.
- This gives an interpretable test of non-competitive behaviour without estimating the entire dynamic game.

Interpretation: The incremental value is a reduced-form test with formal game-theoretic foundations. It keeps the transparency of a screen while addressing the main economic objection to descriptive red flags.

2.3 Contribution relative to regression discontinuity methods

- The forcing variable is not a policy threshold or administrative cutoff. It is an endogenous bid difference generated by strategic auction behaviour.
- The paper provides conditions under which this endogenous running variable can still generate an as-if-random comparison near zero.
- The proof relies on smooth residual demand and then extends the argument to dynamic Markov environments through competitively enforced equilibria.

Interpretation: The paper extends the intuition of RD designs to a strategic setting in which the running variable is chosen by bidders, but local comparison remains informative under competition.

2.4 Contribution for enforcement and policy

- The test is simple enough for procurement agencies, compliance officers, and antitrust authorities to implement using standard bid data.
- It can be applied to backlog, incumbency, geography, subcontracting, joint bidding, or any predetermined covariate predicted by a theory of collusive allocation.
- The authors emphasize that the test is a screen, not a substitute for direct evidence. A rejection indicates non-competitive behaviour or a failure of the competitive null, but not necessarily a legally prosecutable cartel.

Interpretation: The incremental contribution is a practical bridge between academic IO theory and real-world cartel screening.

3 Framework and data measurement

3.1 Dynamic procurement game

- In each period t , a buyer procures one item from a finite set of potential suppliers N .
- The contract is allocated through a sealed-bid first-price auction with public reserve price normalized to one.
- Each bidder chooses whether to participate and, if participating, submits a bid $b_{i,t} \in [0, 1]$.
- Participating is costly: submitting an actual bid requires a cost $k > 0$.
- The lowest bid wins, with ties broken uniformly.

Interpretation: The participation cost matters theoretically. It helps rule out competitive equilibria with non-smooth residual demand and supports the close-bid randomization result.

3.2 State transitions and endogenous dynamics

- Before bidding, bidders observe an industry state θ_t .
- The state follows an endogenous Markov process: the distribution of θ_{t+1} depends on the previous state and the identity of the winning bidder.
- The state can include backlog, recent wins, incumbency, project type, location, bidder capacity, or unobserved heterogeneity.
- Because winning affects future states, even a competitive dynamic environment can generate bid rotation or incumbency patterns.

Interpretation: This is why a simple correlation between wins and backlog or wins and incumbency is not enough to infer collusion.

3.3 Information and costs

- In addition to θ_t , each bidder observes a private signal $z_{i,t}$.
- Signals may be arbitrarily correlated across bidders.
- Costs are private values: bidder i observes its own cost $c_{i,t}$.
- Conditional on the state and bidder i 's signal, bidder i 's cost gives no additional information about other bidders' costs.

Interpretation: The framework is broad enough to include asymmetric independent private values, correlated signals, and unobserved cost heterogeneity.

3.4 Econometric observables

- The econometrician observes bidder-level characteristics $x_{i,t}$ before bidding.
- In the applications, $x_{i,t}$ is backlog or incumbency status.
- The observable $x_{i,t}$ can be a noisy or coarse measure of the true state θ_t .
- The theoretical result allows $x_{i,t}$ to be imperfectly measured, as long as it is predetermined at the time of bidding.

Interpretation: This is crucial in practice because regulators rarely observe the exact cost state that bidders use.

3.5 Ohio school milk auctions

- The first empirical application re-examines the Ohio school milk auctions studied by Porter and Zona.
- The data include auctions in which some firms were later implicated in collusion and a control sample of auctions with no defendant firms.
- The key covariate is incumbency: whether a bidder won the district’s auction in the previous year.
- The analysis compares close winners and close losers separately in the non-competitive and control samples.

Interpretation: The Ohio data provide a validation setting because the identity of colluding firms is externally known from earlier work.

3.6 Japanese municipal procurement auctions

- The second application uses municipal procurement auctions in Japan from 2007 to 2018.
- The total award amount is about 232 billion yen.
- The paper studies backlog and incumbency as possible indicators of non-competitive allocation.
- Backlog is measured using the yen amount of recent work won by a firm over 90-day or 180-day windows.
- Standardized backlog is computed within firm:

$$\bar{x}_{i,t}^{B^k} = \frac{x_{i,t}^{B^k} - \mu_{x_i^{B^k}}}{\sigma_{x_i^{B^k}}},$$

where $k \in \{90, 180\}$.

- Incumbency is defined by whether a firm previously won an auction with the same project name in the same municipality.

Interpretation: Standardizing backlog removes persistent cross-firm differences and focuses on within-firm intertemporal variation in workload.

4 Empirical strategy and theory

4.1 Running variable and discontinuity

The running variable is the difference between bidder i ’s bid and the best competing bid:

$$\Delta_{i,t} = b_{i,t} - \wedge b_{-i,t}. \quad (1)$$

- $\Delta_{i,t} < 0$: bidder i wins.
- $\Delta_{i,t} > 0$: bidder i loses.
- $\Delta_{i,t} \approx 0$: bidder i is a marginal winner or marginal loser.

The discontinuity in a predetermined bidder covariate $x_{i,t}$ is

$$\beta = \lim_{\Delta \downarrow 0} E[x_{i,t} \mid \Delta_{i,t} = \Delta] - \lim_{\Delta \uparrow 0} E[x_{i,t} \mid \Delta_{i,t} = \Delta]. \quad (2)$$

Interpretation: The test is about whether close losers and close winners have the same predetermined characteristics.

4.2 Predictions under collusive allocation

- If a cartel rotates winning opportunities toward firms with low backlog, close winners should have lower backlog than close losers. Since β is loser minus winner, the prediction is $\beta > 0$.
- If a cartel protects incumbents, close winners should be more likely to be incumbents than close losers. If x is an incumbent dummy, the prediction is $\beta < 0$.
- The test does not require specifying exactly how the cartel is organized. It only requires choosing a covariate predicted by a theory of collusive allocation.

Interpretation: The same RD framework can test different cartel allocation rules by changing the covariate x .

4.3 Residual demand and smooth-demand intuition

For bidder i with history $h_{i,t}$, define residual demand as

$$D_i(b \mid h_{i,t}) = \Pr(\wedge b_{-i} \geq b \mid h_{i,t}). \quad (3)$$

The probability of winning conditional on submitting a close bid satisfies

$$\Pr(i \text{ wins} \mid h_{i,t}, |b_{i,t} - \wedge b_{-i,t}| \leq \epsilon) = \frac{D_i(b_{i,t} \mid h_{i,t}) - D_i(b_{i,t} + \epsilon \mid h_{i,t})}{D_i(b_{i,t} - \epsilon \mid h_{i,t}) - D_i(b_{i,t} + \epsilon \mid h_{i,t})}. \quad (4)$$

If D_i is smooth and strictly decreasing, then for small ϵ this probability is approximately $1/2$.

Interpretation: Conditional on a nearly tied bid, winning or losing should be determined by an arbitrarily small bid perturbation, not by bidder state.

4.4 Corollary: as-if random close bids

The smooth-demand argument implies that for any observable covariate $x \in X$,

$$\Pr(x_{i,t} = x \mid \Delta_{i,t} \in (0, \epsilon)) \approx \Pr(x_{i,t} = x \mid \Delta_{i,t} \in (-\epsilon, 0)). \quad (5)$$

Interpretation: The distribution of predetermined observables must be continuous at the winning threshold under competition.

4.5 Formal foundation: competitively enforced MPE

- Markov perfect equilibrium alone does not rule out collusive equilibria when the state evolves endogenously.
- The authors define bidding behaviour as *sensitive* when bidders' expected profits change with their bids at some history.
- An equilibrium is *competitively enforced* if bidders' continuation values at the end of an auction do not reward collusive allocation patterns.
- Under competitively enforced MPE, equilibrium beliefs conditional on close bids satisfy the as-if random property.

Interpretation: This section shows that the RD test is not just heuristic. It has formal foundations in a dynamic game when equilibrium incentives are competitive rather than collusive.

4.6 Empirical estimator

The paper estimates the discontinuity using local linear regression around $\Delta_{i,t} = 0$:

$$\hat{\beta} = \hat{b}_0^+ - \hat{b}_0^-, \quad (6)$$

where $\hat{b}_0^+$ and $\hat{b}_0^-$ are the fitted intercepts to the right and left of zero.

- The paper uses a triangular kernel.
- Bandwidths are selected using the Calonico–Cattaneo–Titiunik procedure.
- Standard errors are clustered at the auction level.
- In the Japan analysis, bids are restricted to cases with $|\Delta|$ less than 20% of the reserve price to avoid misrecorded distant bids.

Interpretation: The local linear RD focuses precisely on near ties, where the competitive null has the sharpest implication.

5 Identification and interpretation

5.1 Why the RD comparison removes cost-based explanations

- Backlog and incumbency can affect costs under competition.
- However, close winners and close losers bid nearly the same price.
- If bidding is competitive and close-bid assignment is as-if random, cost-relevant covariates should not jump at $\Delta = 0$.
- A discontinuity therefore suggests that the allocation rule is not merely reflecting smooth cost heterogeneity.

Interpretation: The test is designed to solve the central problem in interpreting bid rotation and incumbency advantage.

5.2 What rejection means

- A rejection of $\beta = 0$ rejects the competitive null under the maintained assumptions.
- It does not by itself prove a legally actionable cartel.
- It can reflect collusion, non-Markov strategies, or other non-competitive behaviour.
- Direct evidence is still needed for prosecution, but the screen helps identify where to investigate.

Interpretation: The paper is careful to position the method as a screen, not as a substitute for legal evidence.

5.3 Why high-bid and low-bid samples matter in Japan

- The authors split Japanese bids by whether they are above or below the municipality-specific median winning bid.
- Above-median bids are more suspicious because elevated bids are more consistent with reduced competition.
- The discontinuities in backlog and incumbency are concentrated in the above-median sample.
- This strengthens the interpretation that the RD patterns reflect non-competitive behaviour.

Interpretation: The split acts as a diagnostic: the screen has stronger power where collusion is more plausible.

5.4 Placebo test

For non-winning bidders, the paper defines $\Delta_{i,t}^2$ as the difference between bidder i 's bid and the second-lowest bid among losing bidders. Under cartel allocation, the order of losing bidders should be less important than the identity of the winner.

Interpretation: The placebo checks whether discontinuities are uniquely tied to the winner/loser threshold rather than general sorting among losing bids.

6 Tables and figures

6.1 Figure 1: motivating discontinuity in Japanese backlog

Read:

- Figure 1 plots 90-day standardized backlog against bid difference Δ in Japanese municipal auctions.
- Close winners are on the left of zero; close losers are on the right.
- The binned scatter shows a visible jump at zero: marginal losers have higher backlog than marginal winners.

Economic message: This figure visually motivates the RD approach. If close winning were competitive and random, backlog should not jump at the threshold.

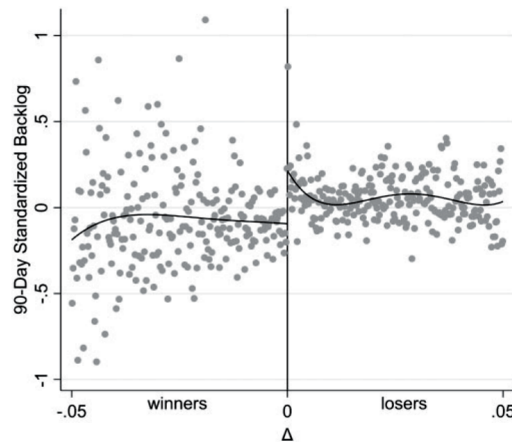


FIGURE 1
Binned scatter plot of standardized backlog, Japanese municipal auctions.

6.2 Tables 1–2 and Figure 2: Ohio school milk sample

Read:

- Table 1 shows that the Ohio sample has 3,754 auctions overall, with 235 non-competitive auctions and 3,267 control auctions.
- The average number of bidders is slightly higher in the non-competitive sample than in the control sample.
- Table 2 shows high incumbency rates in school milk auctions. Incumbents win frequently in both non-competitive and control samples.

- Figure 2 plots the histogram of Δ for the full, non-competitive, and control samples.

Economic message: The high overall incumbency rates show why unconditional incumbency cannot diagnose collusion. The RD test asks whether incumbency jumps exactly at the close-bid threshold.

TABLE 1
Summary statistics of auctions: Ohio school milk auctions

	(1) All All years	(2) Non-competitive All years Excl 83,89		(3) Control All years Excl 83,89	
No. of bidders	1.866 (0.909)	1.983 (0.891)	2.058 (0.882)	1.763 (0.838)	1.770 (0.846)
Winning bid	0.131 (0.013)	0.136 (0.015)	0.138 (0.015)	0.131 (0.013)	0.131 (0.013)
2nd lowest bid	0.135 (0.013)	0.142 (0.015)	0.144 (0.014)	0.135 (0.012)	0.135 (0.013)
3rd lowest bid	0.138 (0.013)	0.147 (0.016)	0.149 (0.014)	0.138 (0.012)	0.137 (0.012)
Obs.	3,754	235	189	3,267	2,658

Notes: The first column corresponds to the set of all auctions, the second column corresponds to the set of auctions in which only the defendant firms bid, and the last column corresponds to those in which no defendant firm bid.

Table 1: Ohio auction summary statistics

defendant firms bid, and the Column (3) corresponds to those in which no defendant firm bid.

TABLE 2
Summary statistics on incumbency: Ohio school milk auctions

	(1) All			(2) Non-Competitive			(3) Control		
	Win/Inc	Ratio	Total	Win/Inc	Ratio	Total	Win/Inc	Ratio	Total
1980	.	.	249	.	.	4	.	.	230
1981	136/185	0.74	273	6/7	0.86	12	123/162	0.76	235
1982	148/188	0.79	287	9/10	0.90	13	131/161	0.81	252
1983	162/214	0.76	318	7/10	0.70	16	150/187	0.80	274
1984	199/249	0.80	339	18/20	0.90	24	174/215	0.81	293
1985	205/260	0.79	357	18/18	1.00	22	177/226	0.78	314
1986	242/293	0.83	378	16/19	0.84	25	216/255	0.85	332
1987	236/287	0.82	411	18/20	0.90	27	211/255	0.83	358
1988	253/304	0.83	419	18/20	0.90	28	227/263	0.86	359
1989	257/332	0.77	392	13/19	0.68	30	236/289	0.82	335
1990	185/247	0.75	331	17/29	0.59	34	165/211	0.78	285
Obs.	3,754			235			3,267		

Table 2: Ohio incumbency statistics

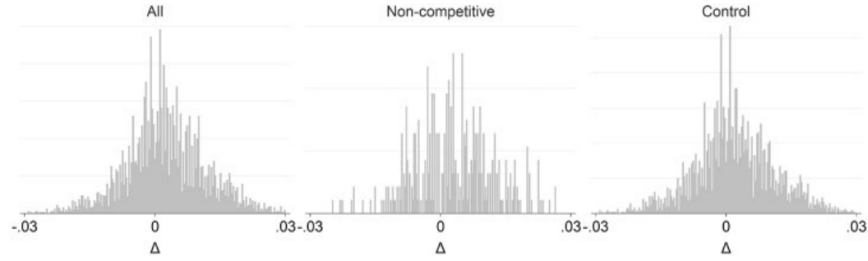


FIGURE 2

Histogram of $\Delta_{i,t}$: Ohio school milk auctions

Notes: The left panel corresponds to the sample of all auctions, the middle corresponds to the sample of non-competitive auctions and the

6.3 Table 3 and Figure 3: Ohio RD results

Read:

- Table 3 reports RD estimates for incumbency in Ohio.
- In the non-competitive sample, the estimated discontinuity is -0.312 for all years and -0.379 excluding 1983 and 1989.
- These estimates mean marginal winners are more likely to be incumbents than marginal losers.
- In the control sample, estimates are small and statistically insignificant: -0.031 and -0.068 .
- Figure 3 shows the binned scatter plots corresponding to the non-competitive and control samples.

Economic message: The test detects the known collusive Ohio sample but does not flag the control sample. This validates the empirical logic.

	(1)	(2)
	Incumbency	
	All years	Exclude 1983 and 1989
Panel (A)		
Non-competitive auctions		
$\hat{\beta}$	-0.312* (0.177)	-0.379** (0.181)
h	0.004	0.005
Obs.	309	266
Panel (B)		
Control		
$\hat{\beta}$	-0.031 (0.063)	-0.068 (0.062)
h	0.004	0.005
Obs.	3,053	2,455

Panel (A) corresponds to the sample of auctions in which only the defendant bidders bid. Panel (B) corresponds to the sample of control auctions in which none of the defendant bidders bid. Standard errors are clustered at the level of the municipality.

Table 3: RD estimates, Ohio auctions

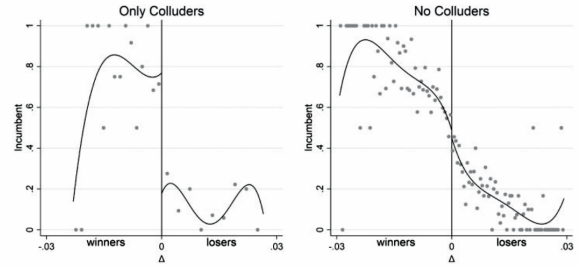


FIGURE 3
Binned scatter plot for incumbency: Ohio school milk auctions

Figure 3: Ohio binned scatter plots

6.4 Table 4 and Figure 4: Japanese auction sample

Read:

- Table 4 reports summary statistics for 11,207 Japanese auctions and 3,377 bidders.
- The average reserve price is about 22.26 million yen, and the average winning bid is about 20.71 million yen.
- The average winning bid is 92.6% of the reserve price.
- The average number of bidders is 6.80.
- Figure 4 shows the distribution of Δ for the full Japanese sample and for bids below or above the municipality median.

Economic message: The Japanese sample is large enough to implement the RD screen and to split bids into more and less suspicious subsets.

	(1) Auctions		(2) Bidders	
	Mean	Std. Dev.	Mean	Std. Dev.
Reserve (Mil. Yen)	22.26	77.14		
Winning bid (Mil. Yen)	20.71	71.78		
Win bid/reserve	0.926	0.083		
No.# of bidders	6.80	4.21		
Incumbent participates (0/1)	0.044	0.204		
No.# of auctions participated			22.56	45.93
No. of wins			3.32	6.97
Raw backlog (90-day)			4.11	17.16
Raw backlog (180-day)			6.45	22.85
Obs.	11,207		3,377	

Table 4: Japanese sample summary statistics

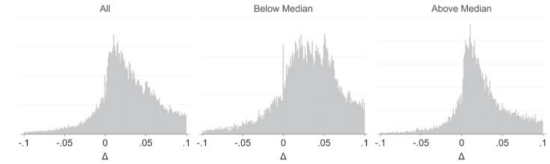


FIGURE 4
Histogram of $\Delta_{i,j}$: municipal auctions from Japan
Notes: The left panel corresponds to the histogram of $\Delta_{i,j}$ for the entire sample. The middle panel corresponds to the sample of bids below the median winning bid of the relevant municipality. The right panel corresponds to the sample of bids above the median. The histogram is truncated at $\Delta_{i,j} = -0.1$ and $\Delta_{i,j} = 0.1$ for readability.

Figure 4: Histogram of Δ in Japan

6.5 Table 5 and Figures 5–6: Japan full-sample RD results

Read:

- Table 5 shows that marginal losers have higher backlog than marginal winners.
- The standardized 90-day backlog discontinuity is 0.136 and statistically significant at the 1% level.
- The raw 90-day backlog estimate is 3.782 million yen and significant at the 10% level.
- The standardized 180-day backlog discontinuity is 0.147 and significant at the 1% level.
- The raw 180-day backlog estimate is 6.747 million yen and significant at the 5% level.
- The incumbency discontinuity is -0.184 , meaning marginal winners are 18.4 percentage points more likely to be incumbents than marginal losers.

- Figures 5 and 6 visualize the backlog and incumbency discontinuities.

Economic message: These patterns match bid rotation and incumbency-based allocation: winners are lower-backlog firms and more likely incumbents.

TABLE 5
Regression discontinuity estimates: municipal auctions from Japan

	(1)	(2)	(3)	(4)	(5)
	90-day backlog		180-day backlog		Incumbent
	Standardized	Raw	Standardized	Raw	
$\hat{\beta}$	0.136*** (0.038)	3.782* (2.250)	0.147*** (0.038)	6.747** (3.157)	-0.184** (0.078)
h	0.020	0.016	0.022	0.015	0.026
Obs.	59,367	63,742	59,413	63,742	2,517

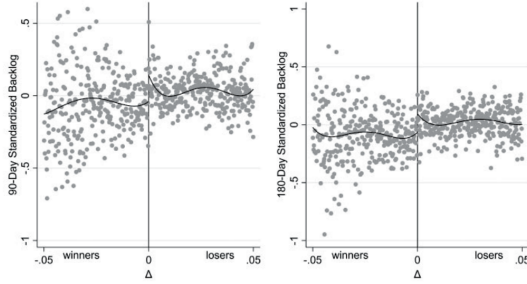


FIGURE 5
Binned scatter plot for 90-day and 180-day standardized backlog: municipal auctions from Japan
note: The curves in the figure correspond to fourth-order (global) nonparametric approximations of the conditional means

Figure 5: standardized backlog

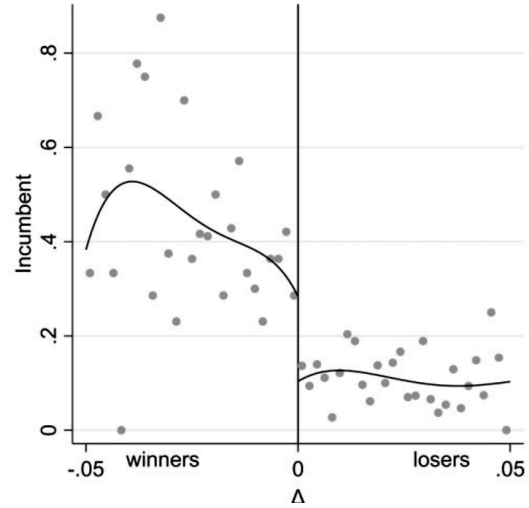


Figure 6: incumbency

6.6 Table 6 and Figures 7–8: high-bid versus low-bid auctions in Japan

Read:

- Table 6 splits bids into above- and below-median winning-bid groups by municipality.
- In above-median bids, the standardized 90-day backlog discontinuity is 0.244, and the raw 90-day backlog discontinuity is 5.572 million yen.
- The standardized 180-day backlog discontinuity is 0.223, and the raw 180-day backlog discontinuity is 9.544 million yen.
- The above-median incumbency discontinuity is -0.260 .
- Below-median bid results are weaker for backlog but still show a negative incumbency discontinuity.
- Figures 7 and 8 show the binned scatter plots for backlog and incumbency in the high- and low-bid samples.

Economic message: The non-competitive pattern is concentrated among elevated bids, which is exactly where collusion is more plausible.

TABLE 6
Regression discontinuity estimates: municipal auctions from Japan

	(1)	(2)	(3)	(4)	(5)
	90-day backlog		180-day backlog		Incumbent
	Standardized	Raw	Standardized	Raw	
Panel (A) : Above median					
$\hat{\beta}$	0.244*** (0.049)	5.572** (2.793)	0.223*** (0.049)	9.544** (3.906)	-0.260** (0.106)
h	0.017	0.016	0.021	0.016	0.034
Obs.	44,945	48,178	44,982	48,178	1,604
Panel (B) : Below median					
$\hat{\beta}$	-0.045 (0.058)	0.377 (2.238)	-0.017 (0.060)	0.108 (3.316)	-0.324** (0.150)
h	0.030	0.024	0.027	0.021	0.026
Obs.	14,438	15,580	14,447	15,580	488

Notes: Panel (A) corresponds to the sample of bids that are above the median winning bid. Panel (B) corresponds to the sample of bids below the median winning bid. Standard errors are clustered at the auction level and reported in parenthesis. The table also reports the bandwidth h used for the estimation. *, **, and ***, respectively denote significance at the 10%, 5%, and 1% levels.

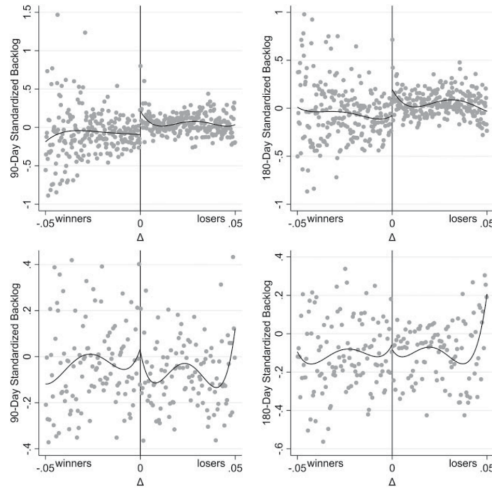


FIGURE 7

Binned scatter plot for 90-day and 180-day standardized backlog: municipal auctions from Japan

Notes: Top panels correspond to Columns (1) and (3) of Panel (A) of Table 6. Bottom panels correspond to Columns (1) and (3) of Panel (B) of Table 6. Left panels correspond to 90-day standardized backlog and the right panels correspond to 180-day standardized backlog. The curves correspond to fourth order (global) polynomial approximations of the conditional means.

Figure 7: high/low bid backlog results

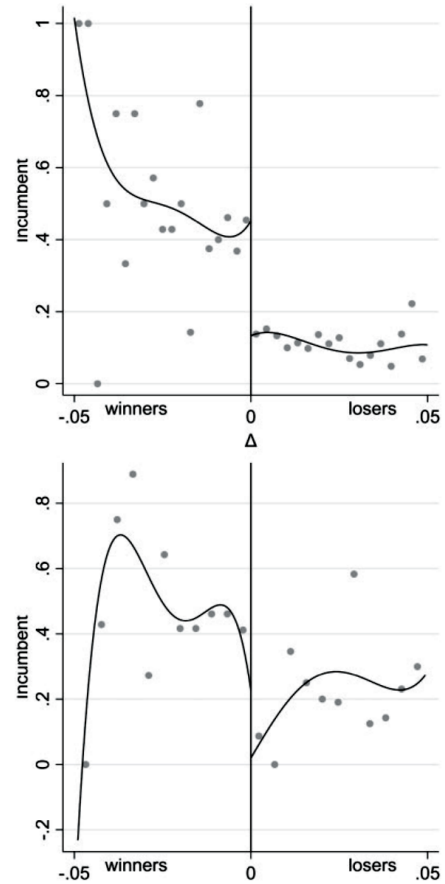


Figure 8: high/low bid incumbency results

6.7 Table 7: placebo test

Read:

- Table 7 uses the difference between a losing bidder's bid and the second-lowest losing bid, Δ^2 .
- The estimates are statistically insignificant at conventional levels.
- The placebo test does not find systematic discontinuities in backlog or incumbency among marginally second and third losing bidders.

Economic message: The discontinuity appears at the winner/loser threshold, not at arbitrary thresholds among losers. This supports the interpretation that the main results capture allocation of winning opportunities.

TABLE 7
Placebo: regression discontinuity estimate with respect to Δ^2

	(1)	(2)	(3)	(4)	(5)
	90-day backlog		180-day backlog		Incumbent
	Standardized	Raw	Standardized	Raw	
Panel (A) : Above median					
$\hat{\beta}$	-0.053 (0.048)	-0.756 (1.433)	-0.016 (0.043)	-1.847 (2.216)	-0.023 (0.058)
h	0.010	0.016	0.013	0.015	0.026
Obs.	41,473	44,802	41,508	44,802	1,466
Panel (B) : Below median					
$\hat{\beta}$	0.035 (0.063)	-0.494 (1.792)	0.026 (0.059)	-0.094 (2.274)	0.080 (0.130)
h	0.025	0.020	0.026	0.020	0.020
Obs.	9795	10,796	9805	10,796	317

Notes: Panel (A) corresponds to the sample of bid that are above the median winning bid. Panel (B) corresponds to the sample of bids that are below the median winning bid. Standard errors are clustered at the auction level and reported in parentheses.

7 Short summary

- The paper develops a regression discontinuity screen for collusion in procurement auctions.
- The method compares close winners and close losers using predetermined bidder covariates.
- Under competition, close winning and losing should be as-if random.
- Under bid rotation, marginal winners should have lower backlog than marginal losers.
- Under incumbent-priority collusion, marginal winners should be more likely to be incumbents.
- The paper provides formal dynamic-game foundations for the test through competitively enforced MPE.
- In Ohio school milk auctions, the method detects known non-competitive auctions but not the control sample.
- In Japanese municipal auctions, close losers have higher backlog and close winners are more likely incumbents.
- The Japanese results are concentrated in above-median bid auctions.
- A placebo test among losing bidders is insignificant.
- The paper offers an implementable antitrust screen that connects observable allocation patterns to non-competitive conduct.

8 Conclusion

8.1 Core conclusion

Kawai, Nakabayashi, Ortner, and Chassang propose a regression discontinuity method for detecting non-competitive behaviour in procurement auctions. The test exploits the idea that close winners and close losers should be statistically similar under competition. If predetermined covariates such as backlog or incumbency jump at the close-bid threshold, the competitive null is rejected.

8.2 Economic intuition

The intuition is that cost-based explanations can explain why low-backlog firms or incumbents win more often on average. But they should not explain why, among bidders who almost tied, winners systematically have lower backlog or higher incumbency. That discontinuity points to an allocation rule beyond smooth competitive cost differences.

8.3 Incremental contribution

The paper’s incremental contribution is to transform familiar cartel red flags into statistically grounded RD tests. It adds formal dynamic-game foundations, works with imperfectly observed covariates, validates the method on known Ohio collusion, and applies it to a large Japanese procurement dataset. The method is practical for regulators because it requires only standard bid data and predetermined bidder characteristics.

8.4 Takeaway

Bid rotation and incumbency advantage are not by themselves proof of collusion. But if close winners and close losers differ discontinuously in backlog or incumbency, the evidence is much stronger. The paper gives regulators a transparent, theory-based way to screen procurement markets for non-competitive behaviour.